PULSEWIRE X — THE NEXT-GEN AI NEWS PORTAL

AI23431 – WEB TECHNOLOGY AND MOBILEAPPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS),

CHENNAI – 602 105

APRIL 2026

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled **" Launch Lens is an AI-powered Startup Showcase and Analytics** " is the Bonafide work of SAQLIN AHAMED (2116241801252) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

Initially I thank the Almighty for being with me through every walk of our life and showering blessings through the endeavor to put forth this report. My sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E, F.I.E.,** our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.,** for providing the requisite infrastructure and sincere endeavoring in educating students in their premier institution.

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SAQLAIN AHAMED – 241801252

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

- PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.
- PEO 2: To prepare students with fundamental knowledge in programming languages, tools and enable them to develop applications.
- PEO 3: To encourage the research abilities and innovative project development in the field of AI, Data Science, networking, security, web development and also emerging technologies for the cause of social benefit.
- PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POS)

- **PO 1: Engineering Knowledge** – Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

- **PO 2: Problem Analysis** – Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

- **PO 3: Design/Development of Solutions** – Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

- **PO 4: Conduct Investigations of Complex Problems** – Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

- **PO 5: Modern Tool Usage** – Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

- **PO 6: The Engineer and Society** – Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

- **PO 7: Environment and Sustainability** – Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

- **PO 8: Ethics** – Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

- **PO 9: Individual and Team Work** – Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

- **PO 10: Communication** – Communicate effectively on complex engineering activities with the engineering community and with society at large.

- **PO 11: Project Management and Finance** – Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

- **PO 12: Life-long Learning** – Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOS)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate:

- **PSO 1: Foundation Skills** – Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, artificial intelligence, data science, machine learning, and networking for solving real-world problems.

- **PSO 2: Problem-Solving Skills** – Ability to apply mathematical methodologies to solve computational tasks, model real-world problems using appropriate AI and data science algorithms, and deliver quality products using open-ended programming environments.

- **PSO 3: Successful Progression** – Ability to apply knowledge in various domains to identify research gaps and provide solutions to new ideas, inculcate passion towards higher studies, creating innovative career paths and evolving as an ethically responsible AI and Data Science professional.

COURSE OBJECTIVES

- To provide foundational knowledge and practical skills in creating and structuring high-performance news portals using HTML5, CSS3, and JavaScript, focusing on accessible and semantic journalism-focused web design.
- To implement reactive client-side scripting using Vanilla JavaScript to handle real-time news tickers, dynamic content updates, and interactive data visualizations.
- To master data persistence by utilizing browser local Storage and storage event listeners to simulate a real-time global news distribution system across multiple user sessions.
- To design role-based workflows that facilitate a collaborative editorial pipeline, specifically managing the relationship between investigative journalists and high-level editors.
- To integrate advanced authentication layers, including real-time Phone OTP verification and secure login protocols for administrative control rooms.

COURSE OUTCOMES

On completion of the course, students will be able to:

- CO 1: Design and develop structured, semantic, multi-page web applications with consistent navigation and responsive layouts using HTML5 and CSS3.
- CO 2: Implement reactive dynamic behavior including persistent breaking news banners, real-time AI accuracy meters, and interactive sentiment analysis modules using Vanilla JavaScript.
- CO 3: Apply browser storage strategies to manage a persistent data layer for global emergency broadcasts and local user bookmarks without requiring a page refresh.
- CO 4: Integrate secure authentication workflows for distinct user roles, utilizing simulated OTP verification for journalists and multi-factor authentication for administrative editors.
- CO 5: Design and implement a batch-processing moderation pipeline, allowing editors to simultaneously perform "Deep AI Scans," approve verified reports, or flag misinformation in bulk.

CO-PO-PSO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation: "–"

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	PulseWire X's promotes high-quality journalistic education by enabling aspiring journalists to access advanced tools for AI-powered verification, supporting lifelong learning in digital literacy.
SDG-8	Decent Work and Economic Growth	The platform empowers independent journalists, investigative reporters, and media professionals to share global news and gain verified recognition. It supports economic opportunity through advanced digital editorial tools and role-based career growth.
SDG-9	Industry, Innovation, and Infrastructure	PulseWire X utilizes innovative web technologies, including reactive Vanilla JavaScript, persistent localStorage data layers, and dynamic AI moderation workflows. This builds accessible digital infrastructure for secure and verified information sharing.
SDG-17	Partnerships for the Goals	The multi-user portal fosters deep collaboration between field journalists and administrative editors. By integrating third-party services like for real-time Phone OTP verification, the platform reflects strong technical partnerships for shared global information goals.

ABSTRACT

ABSTRACT

LaunchLens is an AI-powered Startup Showcase and Analytics Platform developed to bridge the gap between startup founders seeking visibility and investors searching for emerging business opportunities. The platform provides a centralized ecosystem where founders can create detailed startup profiles, upload pitch decks, track growth metrics, and gain community validation, while investors can discover, evaluate, and engage with promising startups through intelligent filtering and real-time trend analysis.

The system is built using a modern full-stack architecture with secure authentication, role-based access control (RBAC), real-time database synchronization, and AI-assisted startup analysis. The platform supports both Email/Password and Google Authentication with automatic profile generation for Founder and Investor roles. A secure database layer with Row-Level Security (RLS) ensures protected data access and user isolation.

LaunchLens features a dynamic startup discovery engine with advanced filtering based on industry, funding stage, revenue model, location, and verification status. A real-time momentum scoring system calculates startup popularity using factors such as upvote velocity, engagement activity, and interaction growth, enabling live trending startup rankings. The platform also implements full-text and fuzzy search mechanisms using tsvector indexing and trigram similarity for accurate and efficient startup discovery.

An integrated AI SWOT Analysis module powered by Gemini 2.5 Flash through the Lovable AI Gateway automatically generates structured business insights, including strengths, weaknesses, opportunities, and threats for each startup. Additional functionalities include startup analytics dashboards, milestone tracking, real-time notifications, verified startup badges based on profile completion, and interactive public startup profiles with timelines and comments.

The user interface follows a modern dark-themed SaaS design inspired by professional platforms such as Linear and Vercel, ensuring responsive layouts, intuitive navigation, and an enhanced user experience. LaunchLens successfully demonstrates the implementation of modern web technologies, AI-assisted analytics, cloud database systems, real-time communication, and scalable SaaS architecture within a practical startup ecosystem.

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LIST OF ABBREVIATIONS

PWS	LaunchLens – The Online Startup Showcase
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
OTP	One-Time Password
API	Application Programming Interface
UI	User Interface
CRUD	Create, Read, Update, Delete
CDN	Content Delivery Network
SDG	Sustainable Development Goal
AI	Artificial Intelligence
DS	Data Science
HTTP	HyperText Transfer Protocol
URL	Uniform Resource Locator

CHAPTER 1

INTRODUCTION

1.1 General

The startup ecosystem has rapidly evolved with the growth of digital entrepreneurship, online funding platforms, and technology-driven business innovation. Modern startups require a strong online presence to showcase their products, attract investors, demonstrate market traction, and build credibility within competitive industries. However, many early-stage startups struggle to gain visibility due to the absence of centralized platforms that combine startup showcasing, analytics tracking, investor interaction, and intelligent business evaluation within a single ecosystem.

LaunchLens is an AI-powered Startup Showcase and Analytics Platform designed to bridge the gap between startup founders and potential investors through a modern cloud-based web application. The platform enables founders to create detailed startup profiles, upload pitch decks, track startup performance metrics, and receive AI-generated SWOT analysis, while investors can discover, evaluate, and interact with startups using advanced filtering and real-time trending systems.

1.2 Need for PulseWire X

Many existing startup listing platforms primarily focus only on displaying startup information without providing real-time analytics, intelligent startup ranking, AI-assisted business evaluation, or interactive investor engagement systems. Startup founders often lack tools to monitor growth visibility, audience engagement, and platform traction, while investors struggle to identify promising startups efficiently from large volumes of unstructured data.

Additionally, the project demonstrates that a modern AI-enabled news portal can be developed entirely using standard frontend web technologies without the need for complex server-side programming language — a key learning outcome of the AI23431 course.

1.3 Scope of the System

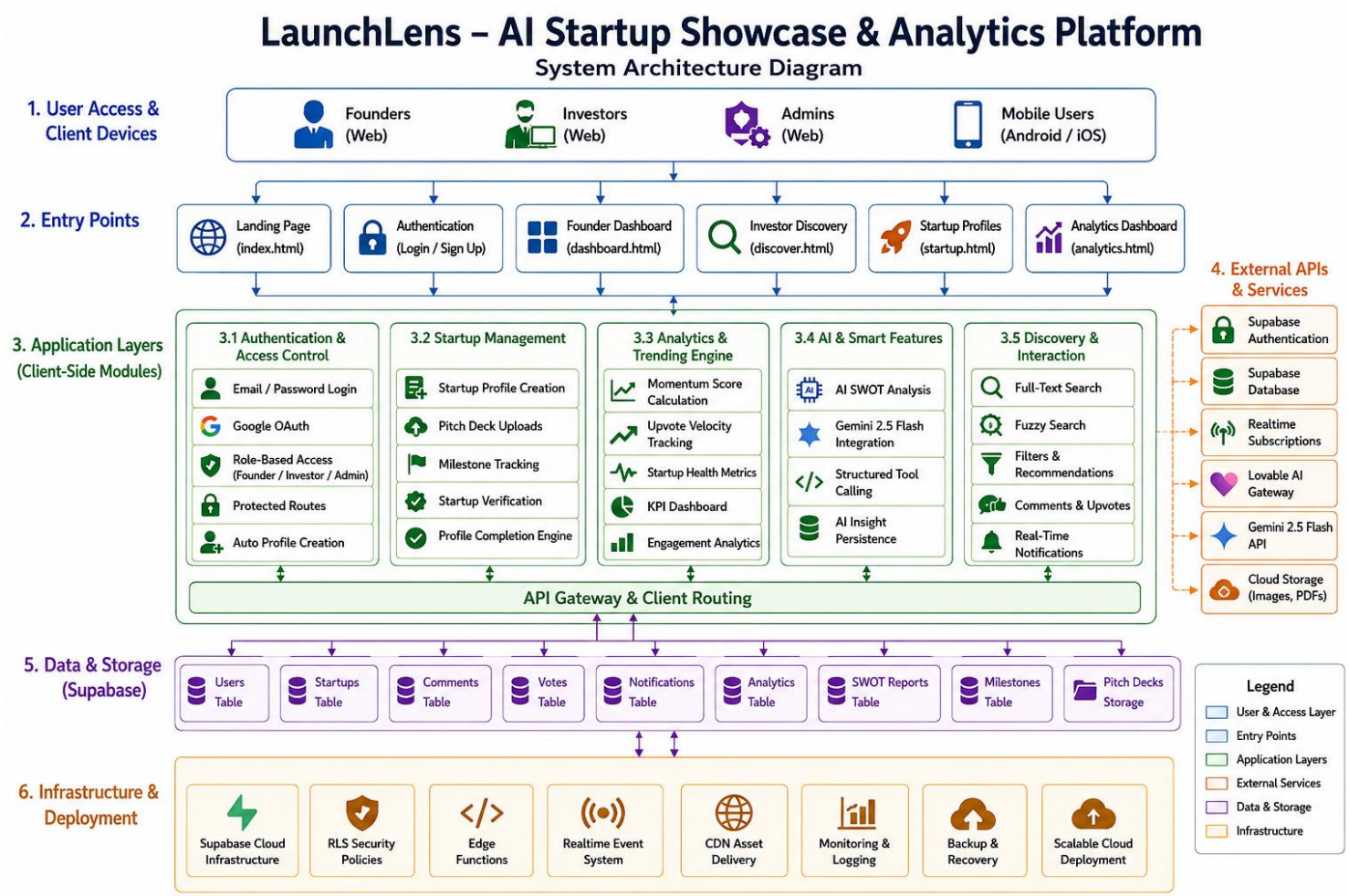
The scope of LaunchLens includes startup profile management, secure founder and investor authentication, startup discovery systems, AI-assisted startup analysis, real-time notifications, and cloud-based analytics tracking. The platform enables founders to create and manage startup profiles with logos, cover images, descriptions-milestones-funding-details-and-pitch-deck-uploads.

1.4 Organization of the Report

This report is organized into seven chapters:

• Chapter 1: Provides the introduction, need, scope, and organization of the LaunchLens platform.
• Chapter 2: Presents the literature survey discussing existing startup showcase systems, investment discovery platforms, and related analytics technologies.
• Chapter 3: Explains the problem formulation and objectives of the proposed system.
• Chapter 4: Describes the system design including architecture, database design, authentication mechanisms, and technical requirements.
• Chapter 5: Discusses the implementation of all major modules including authentication, founder portal, investor discovery, analytics systems, AI SWOT analysis, and real-time communication features.
• Chapter 6: Presents the results, performance evaluation, testing procedures, and system analysis.
• Chapter 7: Concludes the project and discusses future enhancements and scalability improvements

1.5 System Architecture Diagram



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The rapid growth of digital entrepreneurship, startup ecosystems, and online investment communities has transformed how startups gain visibility, attract investors, and present business ideas. Modern startup platforms increasingly rely on real-time analytics, intelligent recommendation systems, cloud-based infrastructure, and AI-assisted business insights to support startup discovery and investor engagement. This chapter reviews existing startup showcase platforms, analytics systems, and AI-powered business evaluation technologies while positioning LaunchLens — AI Startup Showcase & Analytics Platform within the current landscape of modern startup ecosystems.Existing Culinary and Recipe Platforms

Platforms such as Product Hunt, AngelList, Crunchbase, and Y Combinator Startup Directory represent the current generation of startup discovery and investment ecosystems. These platforms provide startup visibility, networking opportunities, funding information, and investor engagement systems. However, many of these systems primarily focus on startup listing and discovery while offering limited real-time analytics, AI-assisted startup evaluation, or interactive engagement tracking for founders.

Social networking platforms such as LinkedIn, X (Twitter), and Reddit have also become important channels for startup promotion and community engagement. However, these platforms lack structured startup analytics, momentum-based ranking systems, intelligent startup filtering, and centralized investor-focused discovery workflows. Startup information is often scattered across multiple platforms, reducing efficiency for both founders and investors.

2.2 Related Work

Research on startup analytics and recommendation systems demonstrates that intelligent ranking algorithms and engagement analysis can significantly improve startup visibility and investor decision-making. Studies on real-time cloud systems highlight the importance of scalable database infrastructure, live notifications, and event-driven architectures for interactive web applications.

Cloud platforms such as Supabase and Firebase demonstrate the effectiveness of modern Backend-as-a-Service (BaaS) solutions for authentication, database management, realtime subscriptions, and secure storage systems in scalable SaaS applications. Research on Role-Based Access Control (RBAC) further supports separating Founder and Investor privileges to improve security, personalization, and workflow management within startup ecosystems.

Recent developments in Generative AI and Large Language Models (LLMs) have enabled intelligent business analysis systems capable of generating structured startup insights, SWOT analysis, and automated business recommendations. AI-powered evaluation systems using models such as Gemini 2.5 Flash improve startup assessment by analyzing startup descriptions, market positioning, strengths, weaknesses, opportunities, and risks dynamically.

Full-text search systems and fuzzy matching algorithms such as trigram similarity and vector indexing have also become important technologies for improving startup discovery accuracy within large-scale digital platforms.

2.3 Summary

The literature survey reveals that while many startup discovery and investment platforms exist, few combine AI-assisted startup evaluation, real-time momentum analytics, advanced startup discovery systems, role-based workflows, cloud-based realtime infrastructure, and intelligent engagement tracking within a single modern SaaS-platform.

LaunchLens — AI Startup Showcase & Analytics Platform addresses this gap by integrating secure authentication, realtime analytics, AI SWOT analysis, startup momentum scoring, advanced filtering systems, and cloud database technologies into a unified startup ecosystem. The platform demonstrates a scalable and practical implementation of modern web technologies, AI-powered analytics, and realtime SaaS architecture suitable for startup discovery, founder engagement, and investor interaction.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Existing startup showcase and investment discovery platforms suffer from several major limitations. First, many platforms only provide basic startup listing functionality without intelligent analytics, realtime engagement tracking, or AI-assisted business evaluation systems. Second, startup founders often struggle to gain visibility and monitor startup performance effectively, while investors face difficulties identifying promising startups from large volumes of unstructured and poorly filtered startup data. Additionally, most platforms lack secure role-based workflows, advanced search systems, realtime trending mechanisms, and centralized startup engagement analytics.

There is a clear need for a modern AI-powered startup showcase platform that demonstrates secure authentication, realtime analytics, role-based access control, intelligent startup discovery, AI-generated business insights, and cloud-based data management — all integrated within a scalable modern web application.

Objectives of the System

The specific objectives of LaunchLens are:

• To implement a secure Founder and Investor authentication system using Email/Password login and Google Authentication with role-based access control.

• To build a startup profile management system where founders can create, edit, and manage startup details including logos, descriptions, funding stages, milestones, and pitch deck uploads.

• To store and manage startup profiles, engagement data, analytics metrics, comments, notifications, and investor interactions using a secure cloud database system.

• To develop a Founder Dashboard with startup analytics, engagement monitoring, momentum tracking, and profile completion systems.

• To provide an Investor Discovery System with advanced filtering, realtime trending startups, full-text search, and fuzzy search functionality.

• To implement AI-assisted SWOT analysis using Gemini 2.5 Flash for generating structured business insights and startup evaluation reports.

• To integrate realtime notifications, comments, and live engagement systems for dynamic startup interaction and communication.

• To deliver a modern SaaS-inspired interface using responsive layouts, realtime dashboards, dark-themed UI design, and interactive analytics visualizations suitable for a next-generation startup ecosystem platform.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design translates the project objectives into a concrete technical architecture. LaunchLens is designed as a modern full-stack SaaS web application where separate modules handle Founder workflows, Investor discovery systems, analytics processing, realtime interactions, and AI-powered business evaluation. The system architecture emphasizes scalability, modularity, and secure cloud-based operations for managing startup profiles, engagement analytics, notifications, and AI-generated insights.System Architecture

The design follows separation of concerns by utilizing frontend technologies for responsive user interfaces, backend services for authentication and database operations, and cloud infrastructure for realtime synchronization and secure data storage. Modern web technologies are used for dynamic functionality, intelligent startup discovery, realtime analytics computation, AI-assisted SWOT analysis, and seamless interaction between founders and investors within the platform.Data Model Design

The LaunchLens database model uses the following table structures:

• Users table: Each record stores {user_id, name, email, role, profile_image, authentication_provider, created_at} — user authentication and role-based access information for Founders and Investors.
• Startups table: Each record stores {startup_name, founder_id, industry, funding_stage, description, logo, cover_image, pitch_deck, location, revenue_model, completion_status, verified_badge, momentum_score, created_at} — complete startup profile data along with analytics and verification status.
• Comments table: Stores investor and community comments linked to public startup profiles.
• Votes table: Stores startup upvotes, engagement activity, and realtime momentum tracking data.
• Notifications table: Stores realtime alerts, engagement notifications, comment updates, and startup interaction events displayed across the platform.
• SWOT Reports table: Stores AI-generated SWOT analysis reports including strengths, weaknesses, opportunities, and threats generated for each startup.
• Milestones table: Stores startup progress updates, product launches, funding achievements, and timeline events associated with startup profiles.

4.2 System Requirements

4.2.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or equivalent / Any ARM processor
RAM	Minimum 2 GB (4 GB recommended)
Storage	Minimum 80 MB for project files; Firebase stores data in cloud

Display	Any screen resolution 1024×768 or above
Internet	Required for Firebase Firestore, Authentication, OTP verification, and CDN resources

4.2.2 Software Requirements

Component	Specification
Operating System	Windows 10+, macOS 10.14+, Ubuntu 20.04+
Browser	Google Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Frontend Technologies	HTML5, CSS3, Vanilla JavaScript (ES6+)
Backend / Database	Google Firebase — database + Authentication
Firebase SDK	Firebase JS SDK v9+ (modular) via CDN
Development IDE	Visual Studio Code (recommended)
Internet Connection	Required at runtime for operations, OTP verification, and real-time updates

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

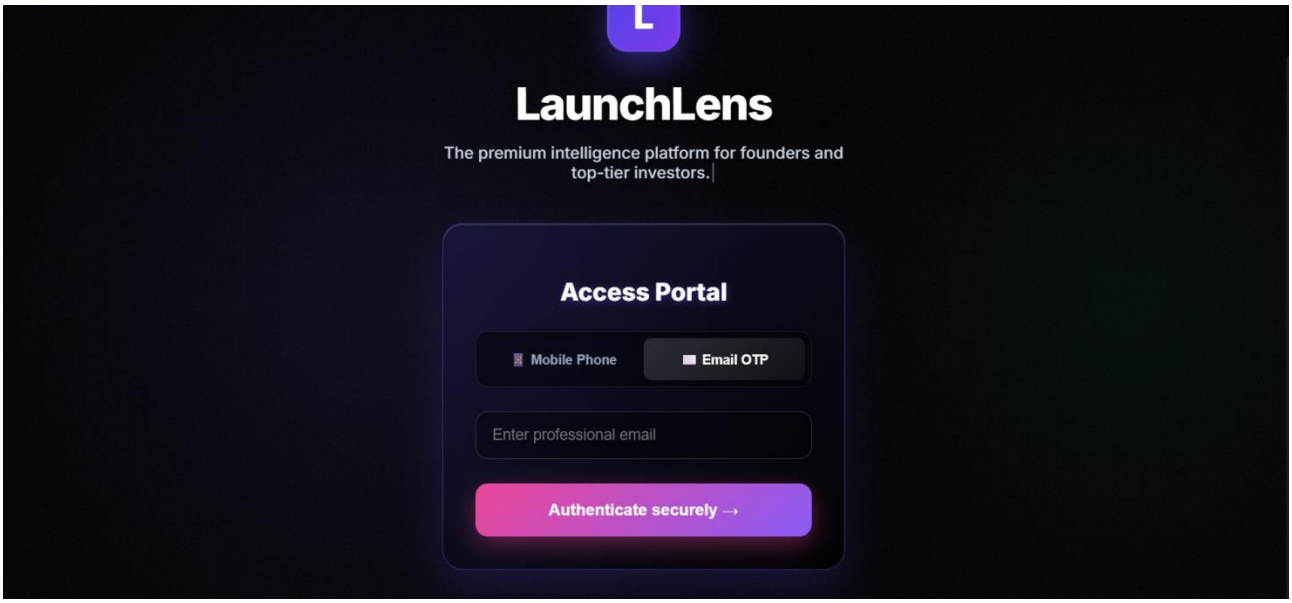
LaunchLens is implemented using a modular full-stack architecture where separate interfaces handle Founder operations, Investor discovery workflows, analytics management, and AI-powered startup evaluation. Development began with project setup — configuring the cloud database, implementing authentication systems, defining Row-Level Security (RLS) policies, and enabling realtime subscriptions for live engagement updates. The frontend modules were then developed in sequence including the landing page, authentication system, founder dashboard, investor discovery platform, analytics dashboard, and AI SWOT analysis integration.

The visual design follows a modern dark-themed SaaS interface inspired by platforms such as Linear and Vercel, featuring responsive layouts, interactive dashboards, smooth animations, and consistent semantic design tokens across all pages. Interactive components such as startup cards, analytics graphs, filters, notifications, comment systems, and momentum indicators follow a clean professional UI aesthetic optimized for startup discovery and investor interaction.

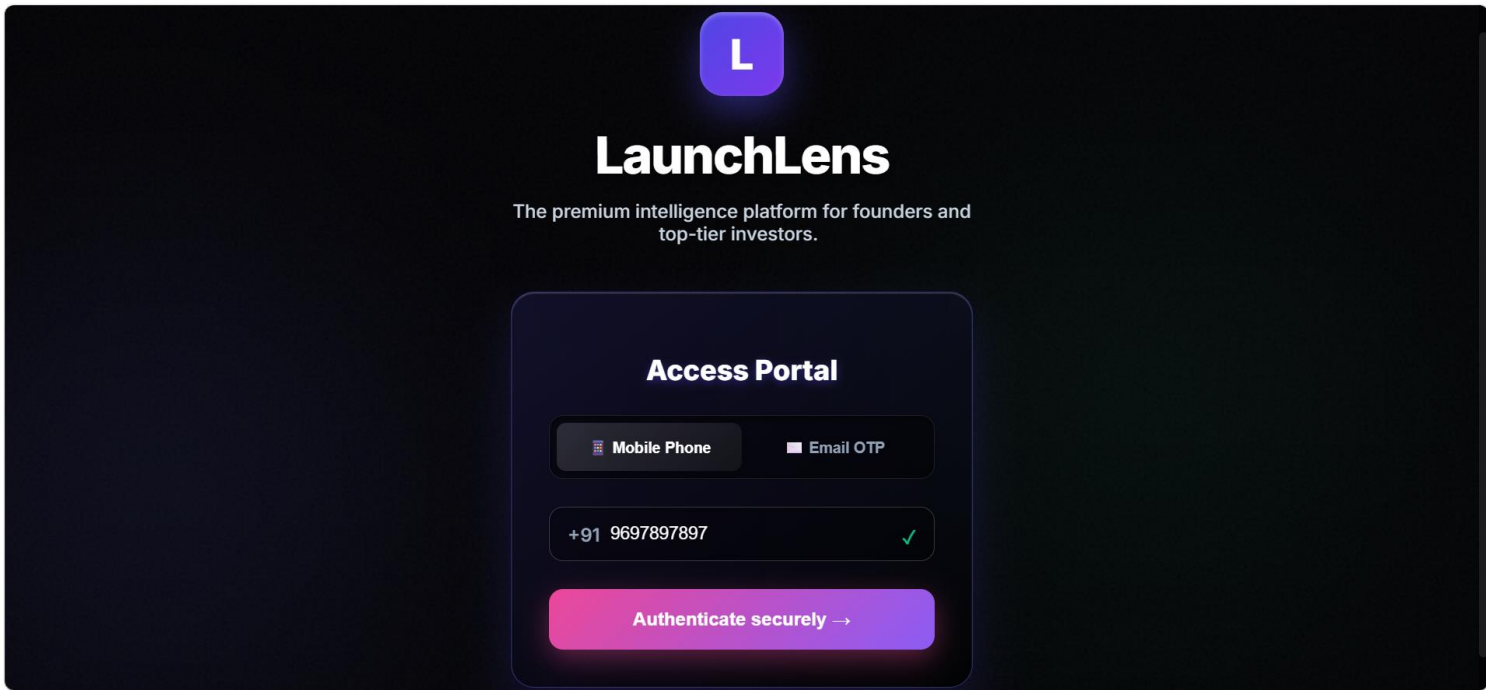
5.2 Module Descriptions

5.2.1 Landing Page / Startup Discovery Module (index.html)

The landing page serves as the main entry point of LaunchLens — AI Startup Showcase and Analytics Platform. It presents the LaunchLens brand identity and modern startup ecosystem theme using a professional dark-themed SaaS interface with interactive startup cards, analytics highlights, and dynamic visual elements. The homepage provides users with direct access to Founder and Investor workflows through-secure-authentication-and-startu-discovery-features.

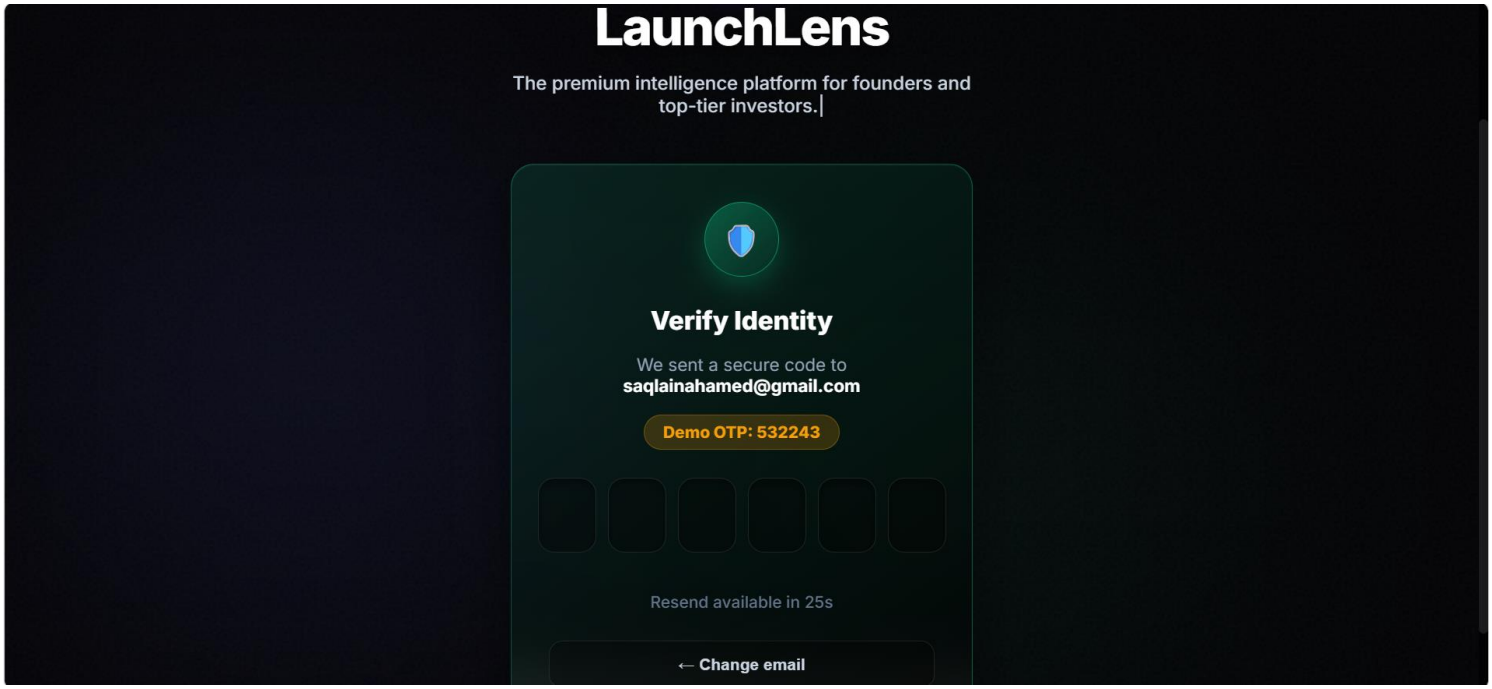


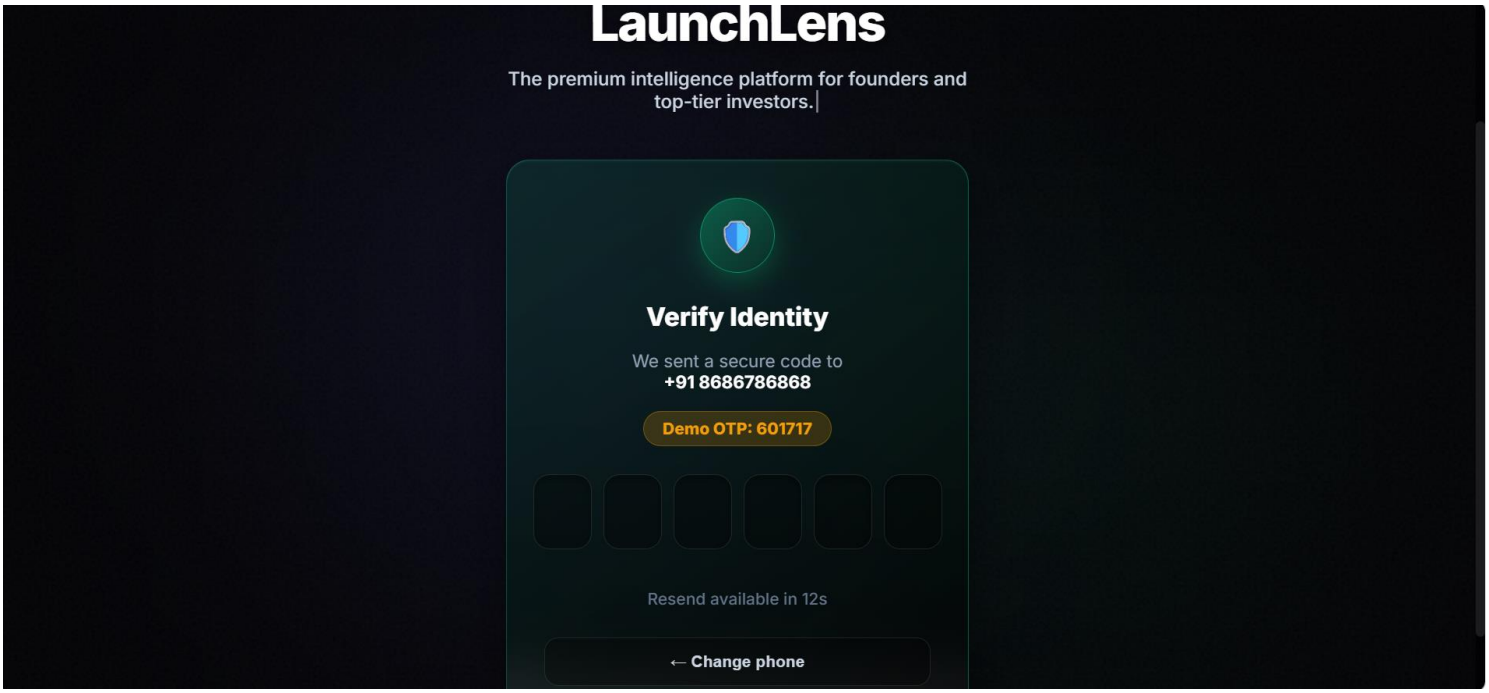
discovery-features.



5.2.2 Founder Authentication Module (auth.html)

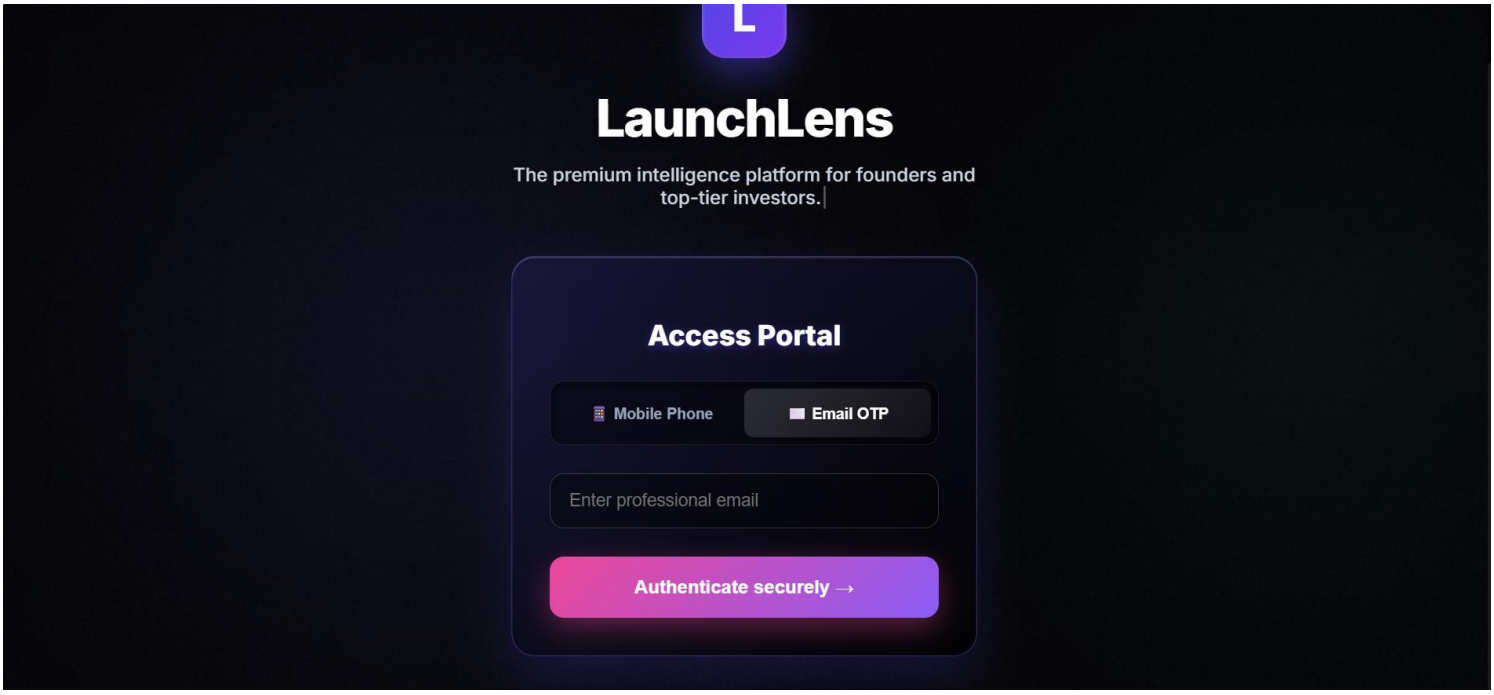
The Editor Authentication page provides a secure entry point for newsroom administrators. It displays a clean login interface with two input fields — editor email/username and password — along with a “Login to Dashboard” button. Upon successful credential verification against the users collection, the editor is redirected to the News Control Panel (admin dashboard) where articles can be created, edited, and published. In case of invalid credentials, an appropriate error message is shown without specifying whether the username or password is incorrect, ensuring security and preventing unauthorized access attempts.

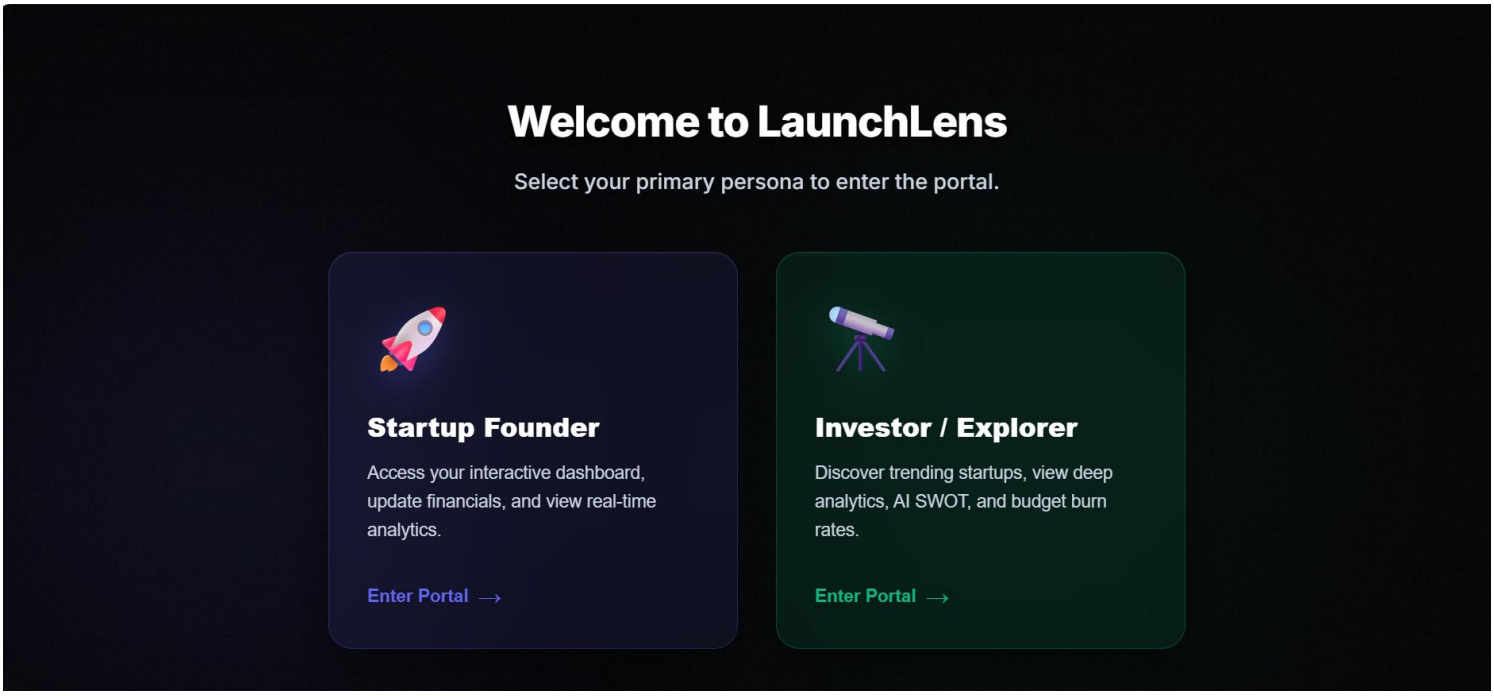

LaunchLens
The premium intelligence platform for founders and top-tier investors.
Verify Identity
We sent a secure code to
saqlainahamed@gmail.com
Demo OTP: 532243
Resend available in 25s
← Change email


LaunchLens
The premium intelligence platform for founders and top-tier investors.
Verify Identity
We sent a secure code to
+91 8686786868
Demo OTP: 601717
Resend available in 12s
← Change phone

5.2.3 Investor Authentication Module (auth.html)

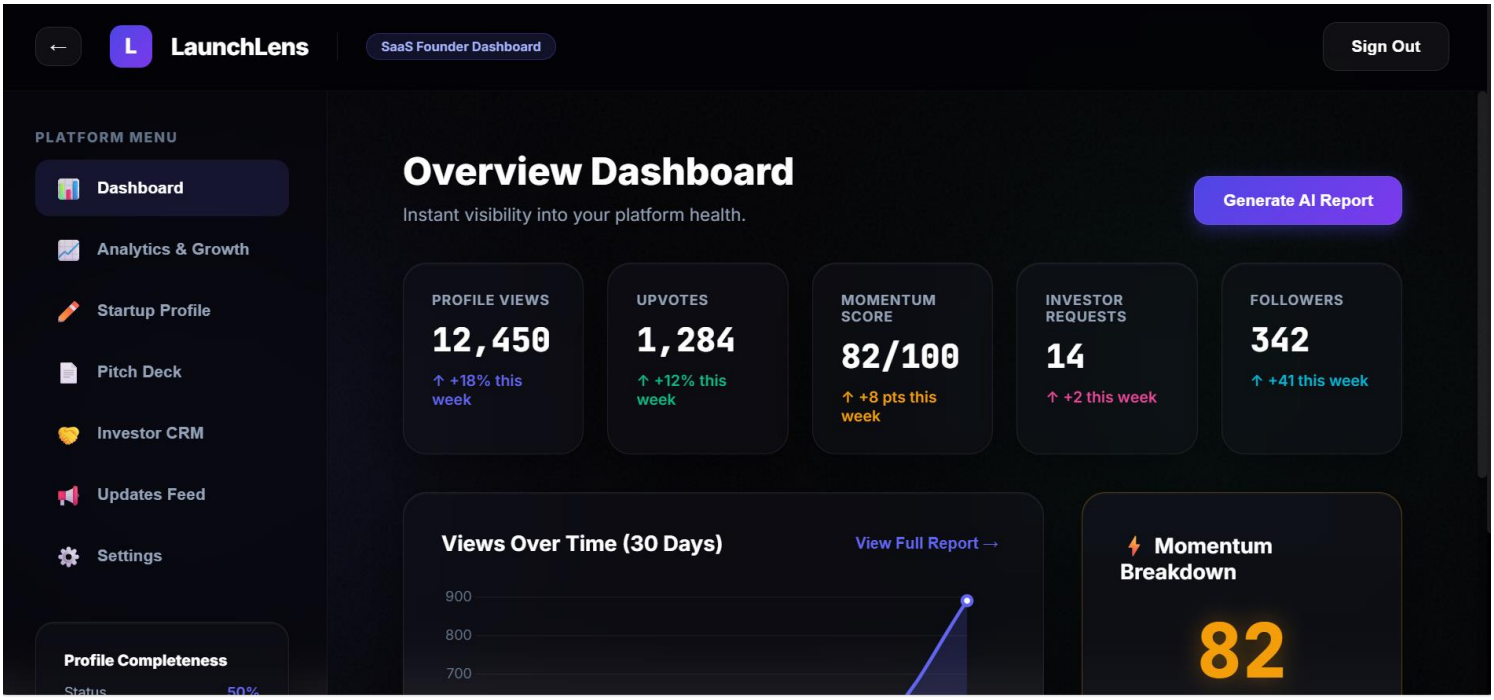
The Investor Authentication module allows investors to securely create accounts and access the LaunchLens platform using Email/Password login or Google Authentication. The registration form enables users to enter details such as full name, email address, investor category, and password. Upon submission, the entered details are validated and securely stored in the users table with the Investor role assigned through the role-based access control system.

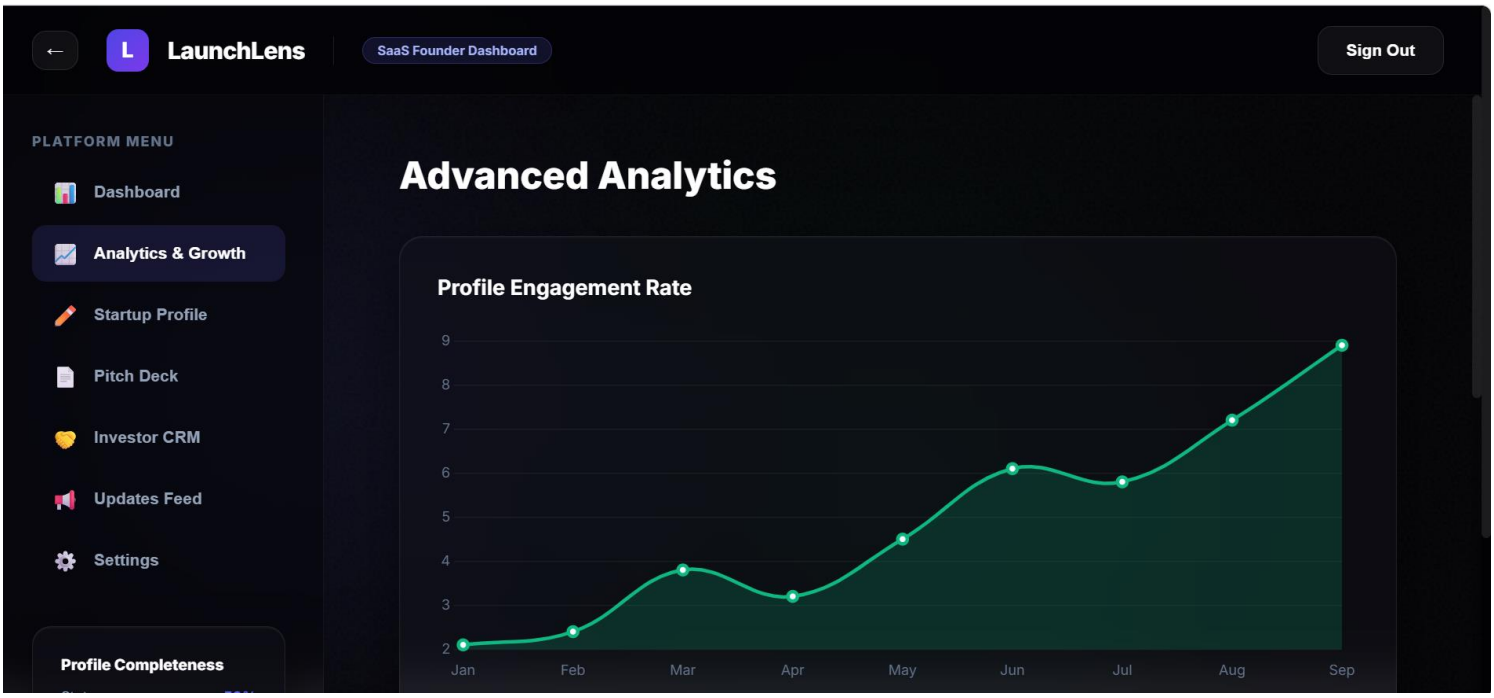




5.2.4 Founder Dashboard Module (dashboard.html)

The Founder Dashboard module allows registered startup founders to securely access and manage their startup ecosystem within LaunchLens. After successful authentication, founders are redirected to the dashboard where they can create startup profiles, upload pitch decks, manage milestones, monitor analytics, and track realtime engagement metrics..

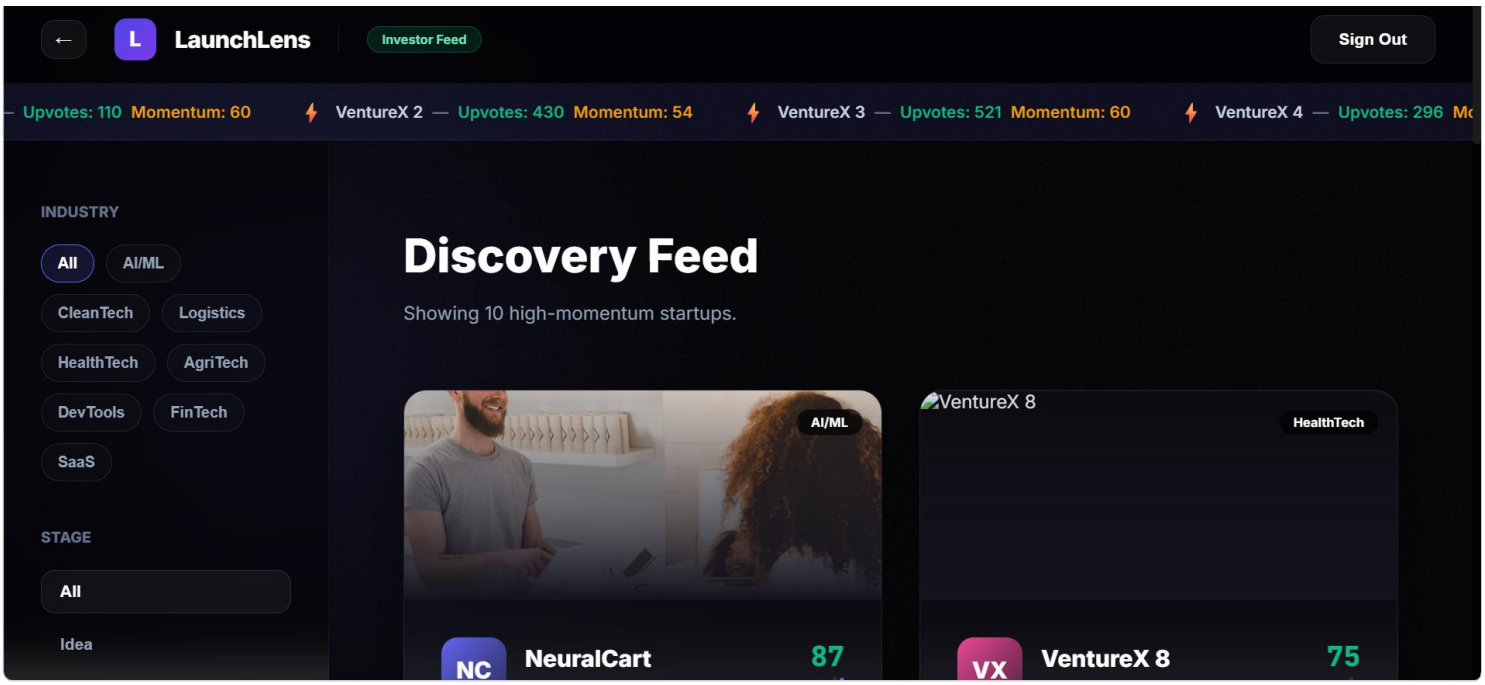

LaunchLens
SaaS Founder Dashboard
Sign Out
PLATFORM MENU
Dashboard
Analytics & Growth
Startup Profile
Pitch Deck
Investor CRM
Updates Feed
Settings
Profile Completeness
Overview Dashboard
Instant visibility into your platform health.
Generate AI Report
PROFILE VIEWS
12,450
↑ +18% this week
UPVOTES
1,284
↑ +12% this week
MOMENTUM SCORE
82/100
↑ +8 pts this week
INVESTOR REQUESTS
14
↑ +2 this week
FOLLOWERS
342
↑ +41 this week
Views Over Time (30 Days)
View Full Report →
900
800
700
Momentum Breakdown
82


LaunchLens
SaaS Founder Dashboard
Sign Out
PLATFORM MENU
Dashboard
Analytics & Growth
Startup Profile
Pitch Deck
Investor CRM
Updates Feed
Settings
Profile Completeness
Advanced Analytics
Profile Engagement Rate
9
8
7
6
5
4
3
2
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep

5.2.5 Startup Profile Creation Module (dashboard.html)

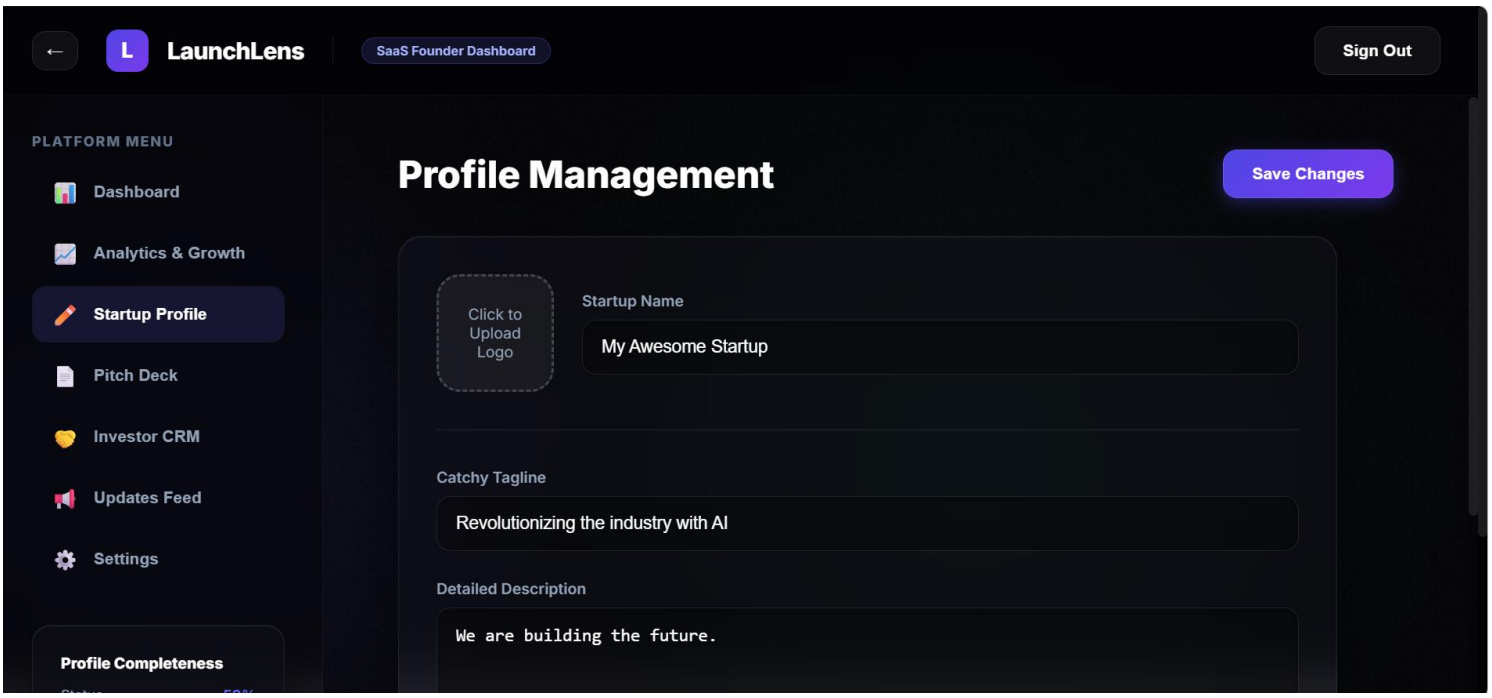
5.2.6

The Startup Profile Creation Module provides founders with an interactive interface to create and manage startup profiles using structured input forms and AI-assisted platform features. It allows founders to enter startup details such as startup name, description, industry, funding stage, location, revenue model, milestones, and social links before publishing the profile to the LaunchLens discovery platform.



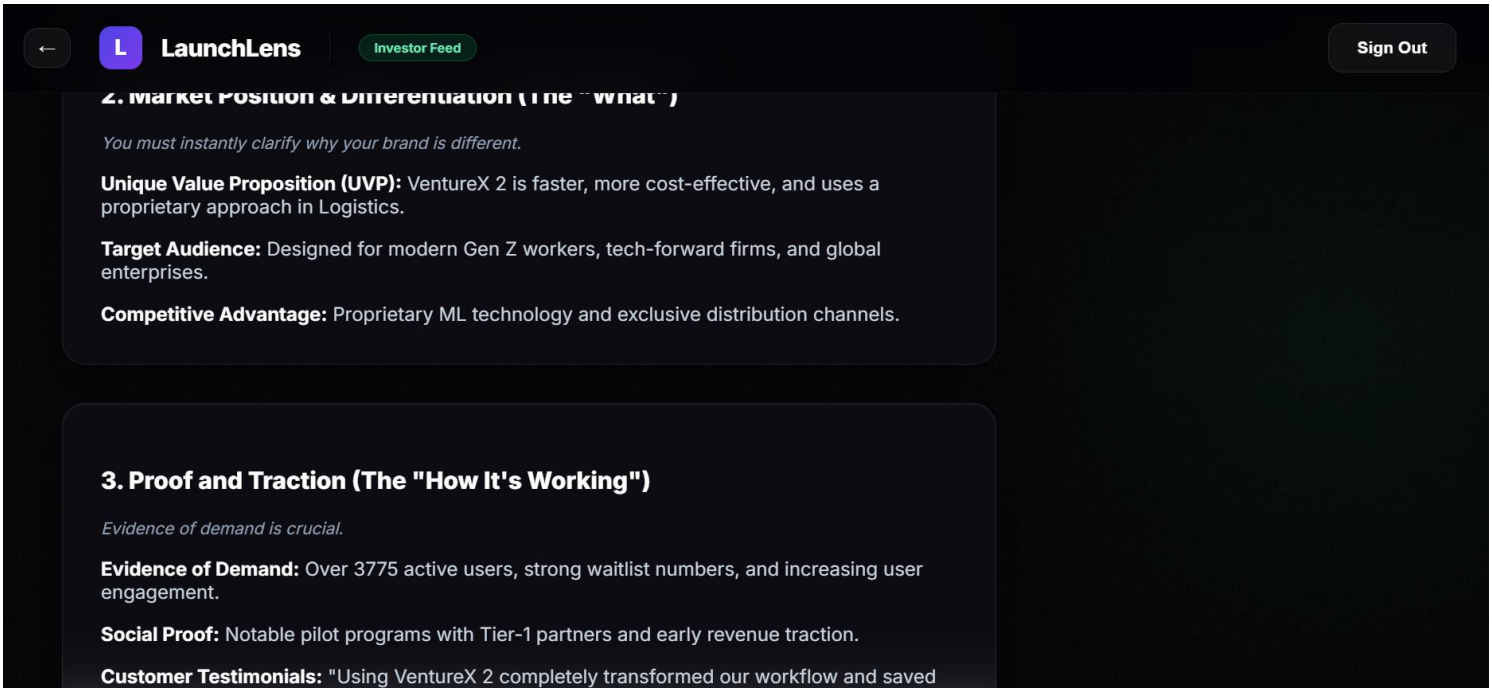
5.2.7 Pitch Deck Upload and Milestone Tracking Module (dashboard.html)

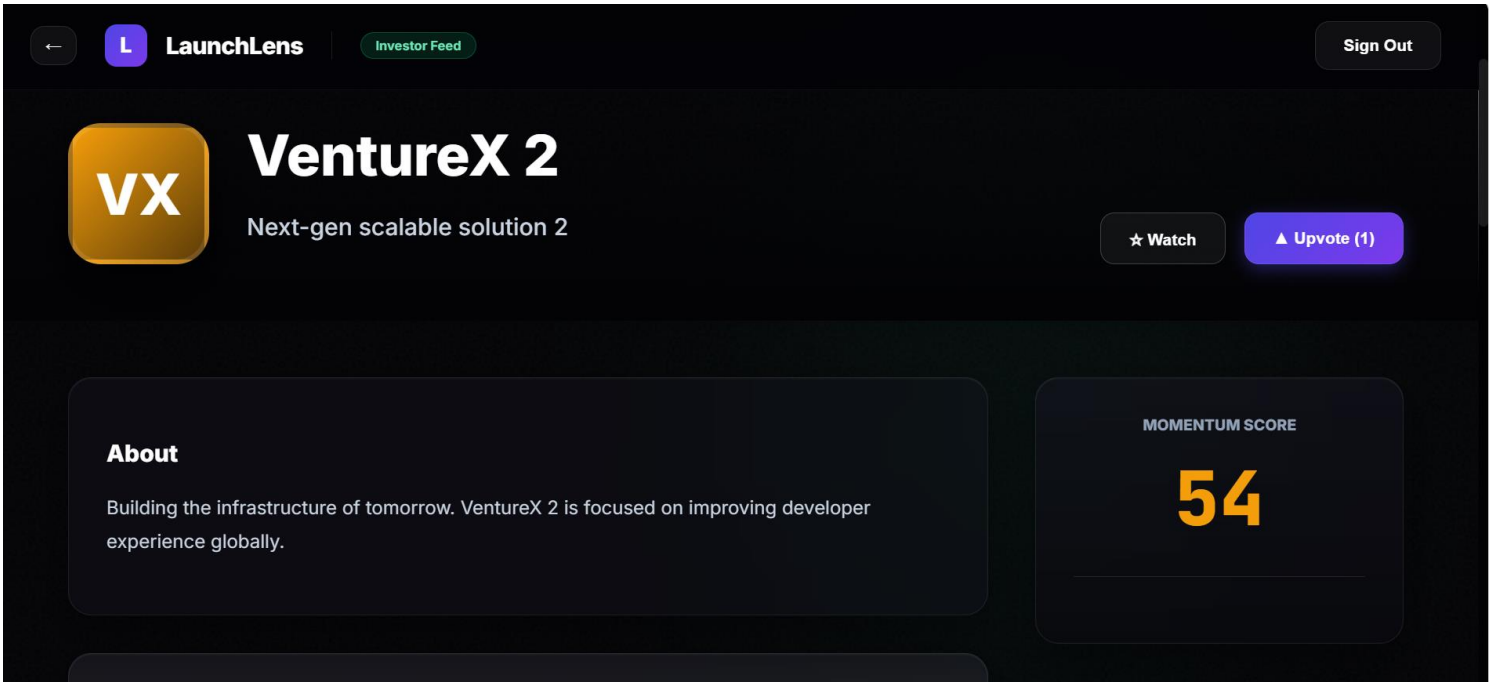
The Pitch Deck Upload and Milestone Tracking Module enables founders to upload startup pitch decks and manage important business milestones directly from the Founder Dashboard. The module allows users to upload presentation files, product documents, and supporting startup materials to improve investor visibility and startup presentation quality



5.2.8 Startup Analytics Dashboard and KPI Monitoring (analytics.html)

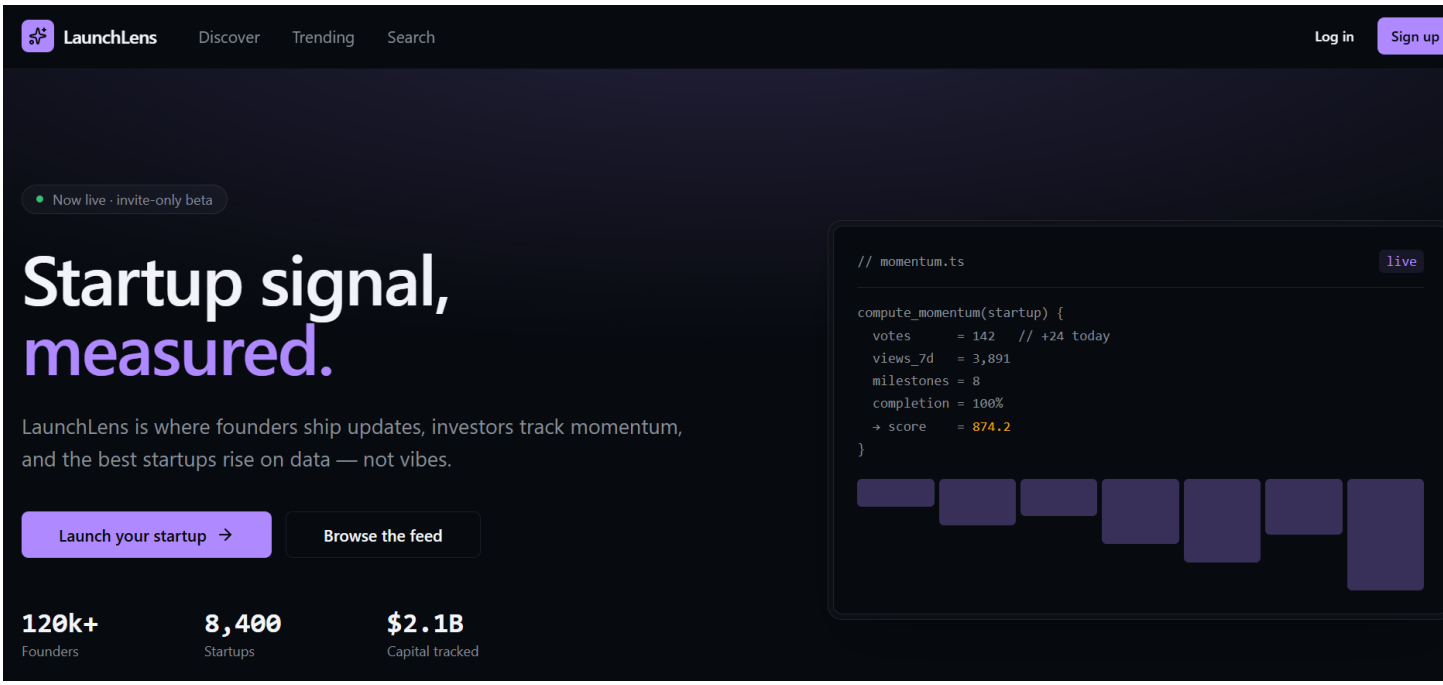
The Startup Analytics Dashboard and KPI Monitoring module helps founders track startup performance and engagement metrics in realtime. It provides detailed analytics related to profile views, investor interactions, upvotes, engagement activity, momentum growth, and startup visibility across the LaunchLens platform. The dashboard enables founders to understand startup traction and monitor audience engagement through interactive charts and performance indicators.

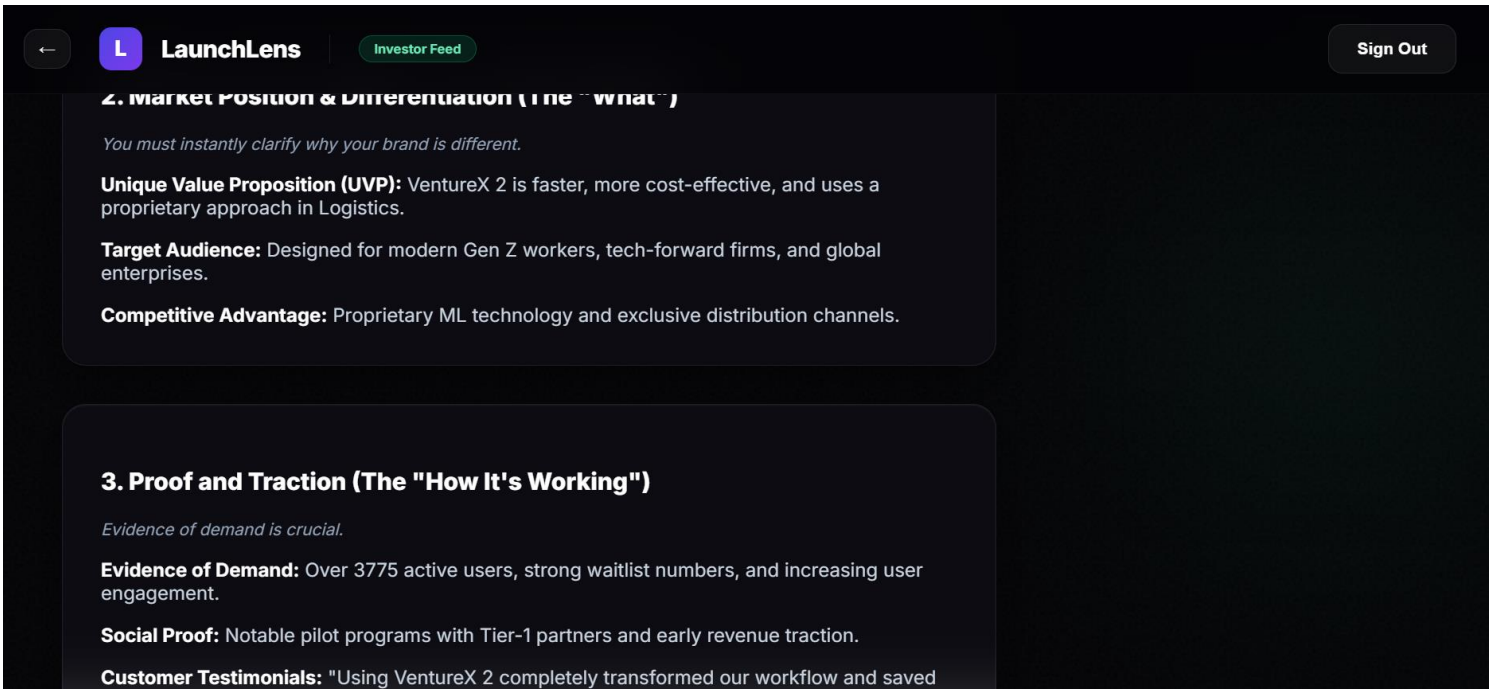




5.2.9 Momentum Scoring and Trending Startups System (trending.html)

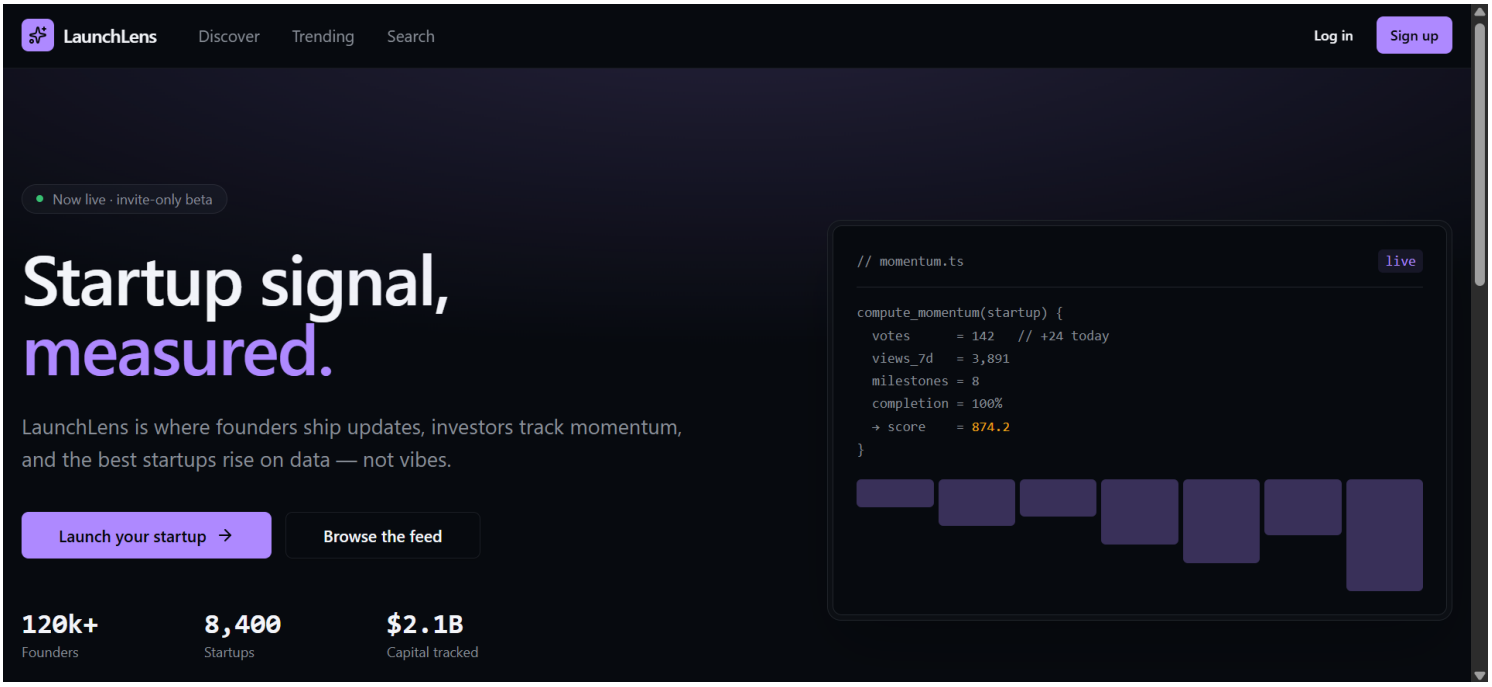
The Momentum Scoring and Trending Startups System evaluates startup popularity and engagement activity using realtime analytics and interaction metrics. The system calculates a dynamic momentum score based on factors such as upvote velocity, profile interactions, comments, engagement growth, and recent activity across the platform. This enables LaunchLens to identify and rank trending startups automatically.





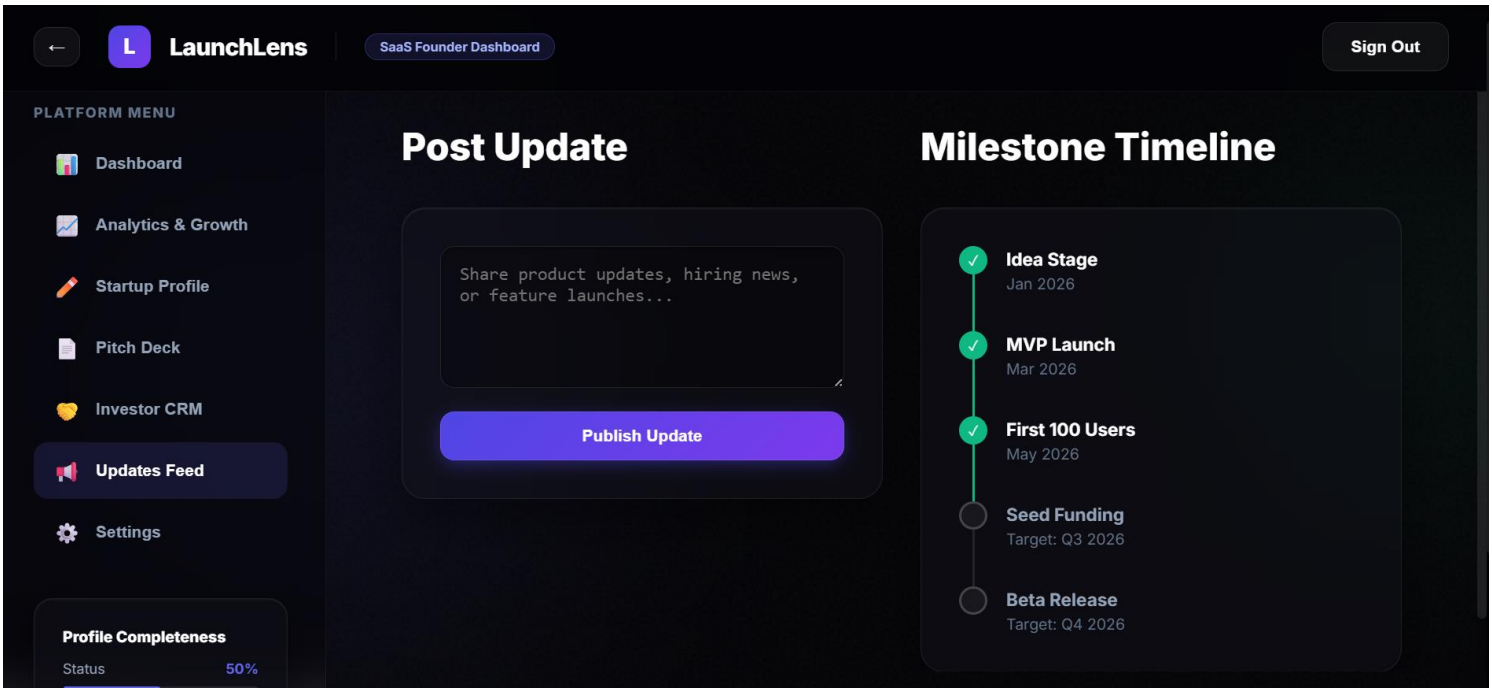
5.2.10 Role-Based Access Control and Verification System

The Role-Based Access Control and Verification System provides secure access management and startup verification functionalities across the LaunchLens platform. The system differentiates users based on Founder and Investor roles, ensuring that each user can access only the features and functionalities authorized for their account type.



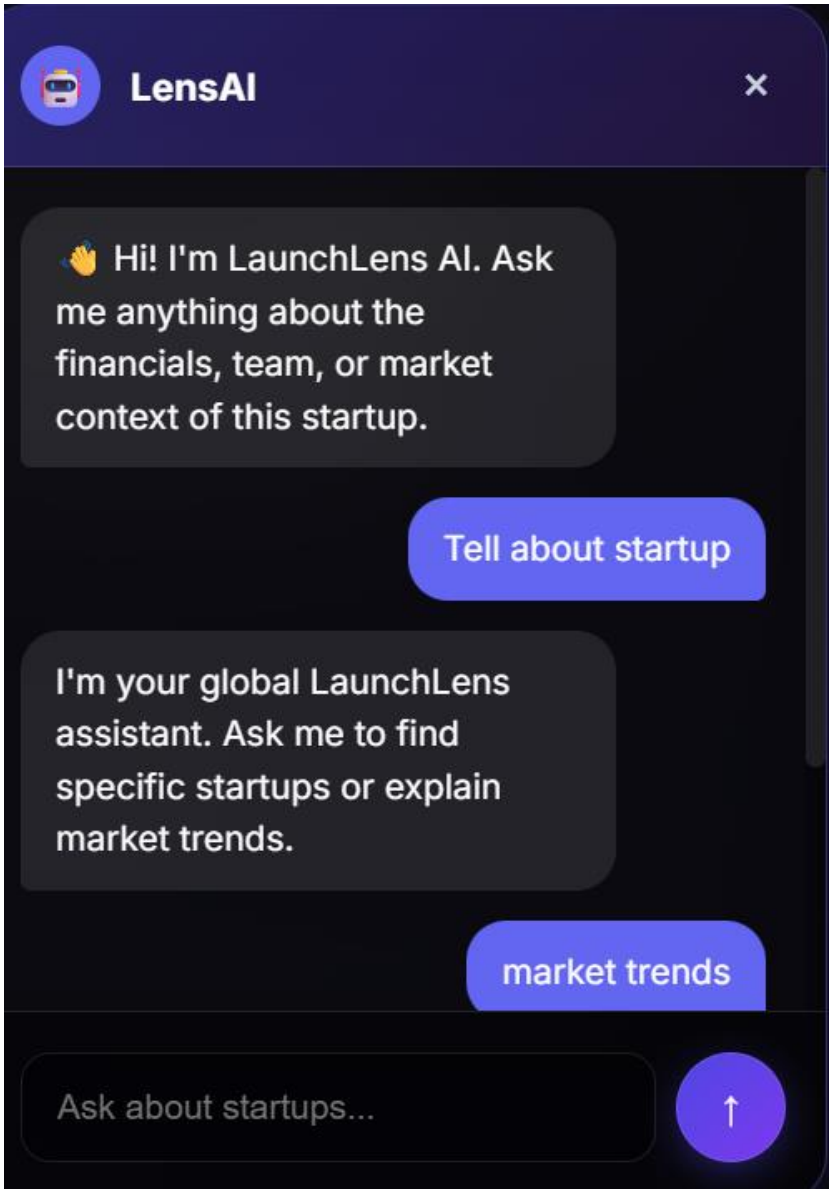
5.2.11 AI SWOT Analysis Assistant Interface

The AI SWOT Analysis Assistant Interface provides intelligent business evaluation capabilities for startups within the LaunchLens platform. The module uses AI-powered analysis to generate structured SWOT reports by identifying the strengths, weaknesses, opportunities, and threats associated with a startup based on its profile information, industry category, and business description.



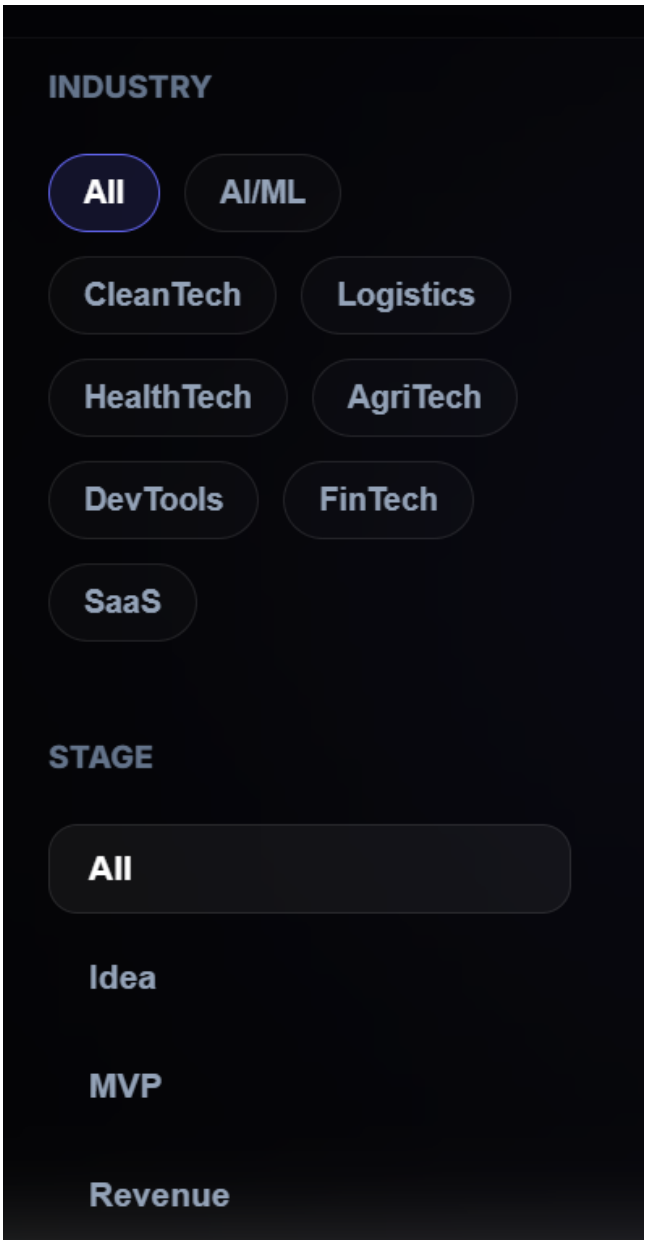
5.2.12 Real-Time Notification and Engagement System (dashboard.html)

The Real-Time Notification and Engagement System provides instant interaction updates and live engagement tracking across the LaunchLens platform. The module notifies founders and investors about important activities such as startup upvotes, comments, profile interactions, verification updates, and momentum score changes-without-requiring manual page refresh.



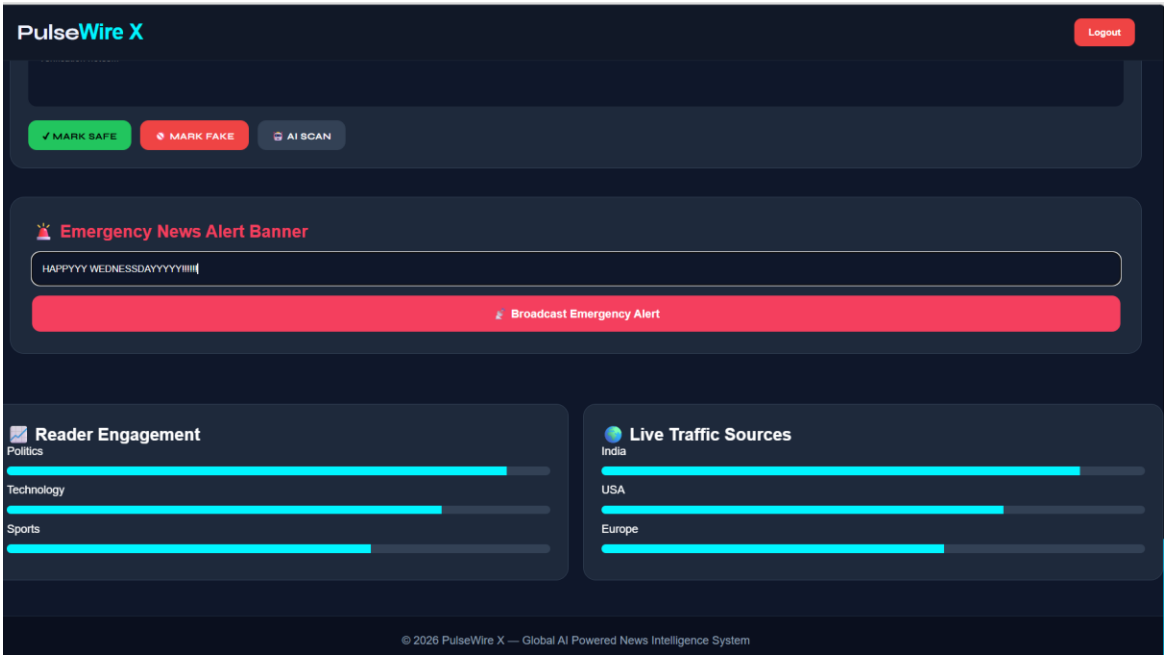
5.2.13 Public Startup Profile and Investor Interaction Module (startup.html)

The Public Startup Profile and Investor Interaction Module provides a dedicated public-facing interface where users and investors can explore detailed startup information within the LaunchLens platform. Each startup profile displays important business details such as startup description, industry category, funding stage, milestones, pitch deck links, momentum score, engagement statistics, and verification status.



5.2.14 Trending Startups Banner and Highlight System (index.html)

The Trending Startups Banner and Highlight System module in the index.html page displays high-momentum startups and important platform highlights at the top of the homepage. It is designed to immediately attract user attention using dynamic startup cards, animated banners, highlighted statistics, and realtime momentum indicators integrated within the LaunchLens interface.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The LaunchLens platform was successfully developed and tested across Chrome, Firefox, and Edge browsers on Windows systems. All core modules — landing page, Founder and Investor authentication, startup profile management, analytics dashboard, startup discovery system, and AI SWOT analysis interface — rendered correctly and operated as intended. The platform successfully stores user accounts, startup profiles, pitch decks, milestones, comments, notifications, analytics data, and engagement records, while realtime momentum tracking and startup discovery workflows functioned correctly across the system.

The complete workflow was verified successfully: a founder registered, created a startup profile, uploaded startup details and pitch decks, and the startup became visible on the public discovery platform. Investors successfully explored startup profiles, interacted through comments and upvotes, and realtime momentum scores updated dynamically on the Trending Startups section. The database tables confirmed proper storage and retrieval of all startup, analytics, engagement, and notification records throughout the platform.

6.2 Performance Evaluation

Since LaunchLens uses cloud-based backend infrastructure with realtime database services and AI integration, system performance primarily depends on response time, internet connectivity, and realtime synchronization efficiency. The following observations were recorded during testing:

• Database reads (fetching startup profiles, analytics data, and trending startups): approximately 300–800 milliseconds on standard broadband connectivity.

• Database writes (startup profile creation, engagement updates, comments, and notifications): under 500 milliseconds with successful realtime synchronization.

• Page load time (including authentication services, analytics rendering, and realtime subscriptions): approximately 1.5–2 seconds on first load; under 700 milliseconds on subsequent loads with browser caching enabled.

• AI SWOT analysis generation using Gemini 2.5 Flash: completed successfully within a few seconds depending on startup profile complexity and network conditions.

• Realtime notifications, momentum score updates, and engagement synchronization operated instantly without requiring manual page refresh.

The system performed reliably throughout all functional test cases with no critical errors, synchronization failures, or realtime communication issues encountered during testing.

6.3 System Testing

Systematic test cases were designed to cover all major user workflows, realtime interactions, analytics systems, and edge cases within the LaunchLens platform. The following table summarizes the results

TC#	Test Case	Input / Action	Expected Output	Result
1	Founder Registration	Enter name, email, password → Register account	Founder account created successfully	PASS
2	Founder Login	Enter email and password → Login to Dashboard	Founder Dashboard opens successfully	PASS
3	Investor Login	Enter investor credentials → Login	Investor Discovery Dashboard opens successfully	PASS
4	Startup Profile Creation	Enter startup details, industry, funding stage, description → Submit	Startup profile stored successfully in database	PASS
5	Pitch Deck Upload	Upload startup pitch deck and supporting documents	Files uploaded and linked to startup profile successfully	PASS
6	Startup Analytics Dashboard	Open analytics dashboard	KPI metrics, engagement statistics, and momentum data displayed correctly	PASS
7	AI SWOT Analysis	Generate SWOT report for startup profile	AI-generated SWOT analysis displayed successfully	PASS
8	Realtime Momentum Tracking	User upvotes and interacts with startup profile	Momentum score updates instantly on Trending section	PASS
9	Advanced Search and Filtering	Search startups using filters and keywords	Relevant startup profiles displayed accurately	PASS
10	Realtime Notification System	User comments or interacts with startup profile	Instant notification displayed to founder/investor	PASS
11	Public Startup Profile	Open public startup profile page	Startup details, milestones, comments, and analytics displayed correctly	PASS
12	Logout	Click Logout button from dashboard	Session cleared and redirected to landing page	PASS

6.4 Resource Utilisation

LaunchLens architecture is optimized for modern cloud-based SaaS performance, with lightweight frontend modules, responsive UI components, and efficient realtime database operations. The platform utilizes cloud infrastructure for authentication, database storage, realtime subscriptions, analytics synchronization, and AI integration while maintaining smooth performance across all major modules. External assets, AI services, and cloud SDKs are loaded dynamically to reduce initial frontend load and improve application responsiveness.

The platform utilizes scalable backend services with secure database storage, realtime communication systems, and cloud file management for startup profiles, pitch decks, analytics data, notifications, and AI-generated SWOT reports. These resources are sufficient to support realtime startup discovery, engagement tracking, role-based workflows, analytics processing, and investor interactions during project demonstration and testing.

The system architecture is designed to operate efficiently without requiring expensive enterprise infrastructure, making the platform suitable for academic demonstrations, prototype deployment, and scalable future enhancements within modern startup ecosystems.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

LaunchLens successfully demonstrates that a modern AI-powered startup showcase and analytics platform can be developed using advanced web technologies, cloud-based infrastructure, realtime database systems, and AI-assisted business analysis. The project implements all major components of a professional startup ecosystem including secure Founder and Investor authentication, startup profile management, realtime momentum tracking, AI-generated SWOT analysis, advanced startup discovery, and cloud-based engagement systems within a unified SaaS-oriented platform.

The visual design, inspired by modern platforms such as Linear and Vercel, utilizes a professional dark-themed interface with responsive layouts, realtime analytics dashboards, interactive startup cards, and clean SaaS-style UI components to deliver an immersive and modern user experience. All project objectives were successfully achieved and verified through systematic testing of authentication workflows, realtime synchronization systems, analytics processing, and AI integration features.

7.2 Future Enhancements

• Realtime Push Notifications: Implement advanced cloud messaging services to send instant push notifications to founders and investors regarding startup interactions, comments, upvotes, verification updates, and momentum score changes across the platform.
• Direct Pitch Deck and Media Uploads: Replace external file links with secure cloud storage uploads, allowing founders to directly upload pitch decks, startup assets, product demonstrations, and branding materials within the platform.
• Advanced Startup Discovery System: Add intelligent recommendation systems, realtime search suggestions, AI-powered startup matching, and dynamic filtering mechanisms for improved startup exploration and investor targeting.
• Mobile SaaS Application: Develop a React Native or Flutter-based companion mobile application using the same backend infrastructure to extend LaunchLens to Android and iOS platforms for realtime startup engagement and investor access.
• Founder Credibility and Startup Trust System: Add founder reputation profiles displaying startup growth history, engagement analytics, investor trust indicators, verification badges, and platform performance ratings to improve startup credibility and investment confidence.
• AI Investment Recommendation Engine: Integrate machine learning models capable of recommending startups to investors based on engagement history, industry interests, investment patterns, and startup momentum analytics.
• Advanced Analytics and Forecasting: Implement predictive analytics systems to forecast startup growth trends, investor engagement patterns, and future momentum performance using AI-driven business intelligence techniques.

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APPENDICES

Appendix 1: System Pseudocode

System Overview

LaunchLens is a cloud-based startup showcase and analytics platform with secure authentication, realtime database operations, AI-powered SWOT analysis, and startup discovery systems. All startup profiles, analytics data, notifications, comments, and engagement activities are stored securely within the database infrastructure. The platform supports role-based access control for Founders and Investors with realtime synchronization and protected dashboard access.

Founder Registration Logic

```
FUNCTION RegisterFounder(name, email, password)
Validate all fields
Check existing user email
IF email exists
→ Show error message
ELSE
Create founder account
Store user data in users table
Redirect to Dashboard
END IF
END FUNCTION
```

Startup Profile Creation Logic

```
FUNCTION CreateStartup(name, industry, description, fundingStage)
Validate startup details
Create startup record
Store startup data in startups table
Show success notification
END FUNCTION
```

Startup Engagement Logic

```
FUNCTION UpvoteStartup(startupId)
Increase upvote count
Update momentum score
Refresh trending rankings in realtime
Send notification to founder
END FUNCTION
```

AI SWOT Analysis Logic

```
FUNCTION GenerateSWOT(startupId)
Fetch startup details
Generate SWOT analysis using AI
Store SWOT report in database
Display results on dashboard
END FUNCTION
```

Realtime Notification Logic

```
FUNCTION SendNotification(userId, message)
Create notification record
Push realtime notification
Display alert on dashboard
END FUNCTION
```

Appendix 2: System Test Verification Summary

• Total test cases executed: 12

• Test cases passed: 12

• Test cases failed: 0

• Database tables verified: users, startups, comments, notifications, votes, SWOT reports

• Founder workflow: startup profile creation, analytics tracking, and pitch deck uploads verified successfully

• Investor interaction system: comments, upvotes, and realtime engagement tracking verified

• Momentum scoring system: realtime startup ranking and trending updates verified across multiple sessions

• AI SWOT Analysis: successful AI-generated startup analysis verified

• Realtime notification system: instant synchronization confirmed across multiple browser sessions

Final System Status: LaunchLens has been fully developed and successfully tested against a live cloud backend. The system ensures secure role-based access control, realtime startup discovery, AI-powered business analysis, dynamic engagement tracking, and cloud-based analytics processing within a scalable SaaS architecture. The modern dark-themed interface, realtime synchronization systems, and intelligent startup evaluation features confirm the platform's readiness for academic submission and live demonstration.

LinkedIn Learning

HTML, CSS, and JavaScript: Building the Web

Course completed by Saqlain Ahamed

May 12, 2026 at 02:41PM UTC • 3 hours 37 minutes

Top skills covered

Web Development | Cascading Style Sheets (CSS) | HTML

Shea Hanson, Head of Learning Content Strategy

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This professional has demonstrated initiative and a commitment to deepening their skills and advancing their career. Well done!

VERIFIED

12th May 2026

Certificate code : 10218898

Krishna Kumar

CEO, Simplilearn